Unreported RTT Pathway Removals at Mid and South Essex NHS Foundation Trust

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The NHS has published Referral to Treatment (RTT) waiting times data since 2007, based on monthly submissions from organisations providing consultant-led care. Each submission reports the number of new RTT referrals, the number of pathways reaching a clock-stop during the month, and the number of incomplete pathways remaining at month-end. The aggregate of these incomplete pathways is widely reported as the NHS "waiting list".

Taken together, these quantities imply a simple accounting identity: the number of incomplete pathways at the start of the month, plus new referrals, minus completed pathways, should equal the number of incomplete pathways at month-end. In practice, this identity does not hold. The residual is typically negative, indicating a net removal of pathways that is not disaggregated or attributed in the published RTT dataset—commonly described as "unreported removals".

Previous analysis has suggested that such unreported removals average around 3% nationally. In this preliminary analysis, I show that the mean unreported removal rate for Large Acute Trusts, excluding MSEFT, in 2024 was 2.83%. I then examine Mid and South Essex NHS Foundation Trust (MSEFT), which exhibits an unreported removal rate of 8.67% - approximately three times the baseline average—corresponding to around 127,000 pathway removals beyond what would be expected for a comparable trust.

Establishing a Large Acute Trust baseline

As part of the NHS Oversight Framework, NHS England publishes an acute trust league table, which identifies 23 trusts classified as Large Acute Trusts. (NHS England 2025a) The Referral to Treatment (RTT) dataset contains monthly submissions for each of these trusts (NHS England 2025b), segmented by treatment specialty and identified by treatment code. This includes the meta-code C_999, which aggregates activity across all specialties.

Each monthly RTT submission consists of three sections. Part 1 reports completed RTT pathways (clock-stops), subdivided into admitted pathways (Part 1a) and non-admitted pathways (Part 1b). In this analysis, these are collectively referred to as *completed pathways*. Part 2 reports the number of incomplete RTT pathways remaining at the end of the reporting period, referred to here as *incomplete*. Part 3 reports the number of new RTT referrals received during the reporting period, referred to here as *new pathways*.

These quantities imply a simple accounting identity. Absent any unreported changes, the number of incomplete pathways at the end of a reporting period should equal the number of incomplete pathways at the start of the period, plus new pathways, minus completed pathways. Deviations from this identity are calculated as:

$$\text{UnreportedRemovals}_t = \text{Incomplete}_t - \left(\text{Incomplete}_{t-1} + \text{NewPathways}_t - \text{Completed}_t\right)$$

A negative value of this residual indicates a net reduction in incomplete pathways that is not explained by reported referrals or completions in the published dataset.

Methods

Monthly RTT reports were consolidated into a single table and aggregated by provider using the C_999 treatment meta-code. For each period, this yields totals for observed incomplete pathways, expected incomplete pathways implied by reported flows, and the absolute residual between the two. These aggregates are then used to calculate the total number of pathways available to treat and the implied percentage net loss attributable to unreported removals.

In this analysis, values drawn from the publicly released RTT CSV files are referred to as "reported" figures, reflecting their provenance as trust-submitted data published by NHS England. Where these figures enter the accounting identity, they are treated as the "observed" quantities in the statistical sense.

Large Acute Trusts baseline (excluding MSEFT)

Table 1 summarises monthly reported incomplete pathways, expected incomplete pathways implied by reported flows, and the resulting residual for the 22 Large Acute Trusts excluding Mid and South Essex NHS Foundation Trust (MSEFT) during calendar year 2024.

Table 1

	Observed Incomplete Pathways	Expected Incomplete Pathways	Unreported Change	Treatable Pathways	Unreported %
2024-01	1,355,909	1,406,947	-51,038	1,675,204	-3.05%
2024-02	1,356,430	1,406,080	-49,650	1,659,809	-2.99%
2024-03	1,355,242	1,401,626	-46,384	1,649,376	-2.81%
2024-04	1,363,802	1,408,342	-44,540	1,662,020	-2.68%
2024-05	1,371,241	1,417,668	-46,427	1,677,729	-2.77%
2024-06	1,377,303	1,419,740	-42,437	1,670,751	-2.54%
2024-07	1,372,476	1,426,756	-54,280	1,704,923	-3.18%
2024-08	1,368,441	1,415,013	-46,572	1,658,678	-2.81%
2024-09	1,364,129	1,406,322	-42,193	1,671,750	-2.52%
2024-10	1,359,924	1,411,835	-51,911	1,700,623	-3.05%
2024-11	1,347,169	1,395,223	-48,054	1,669,081	-2.88%
2024-12	1,338,080	1,381,081	-43,001	1,621,994	-2.65%

	Observed Incomplete Pathways	Expected Incomplete Pathways	Unreported Change	Treatable Pathways	Unreported %
Mean	1,360,846	1,408,053	−47,207	1,668,495	−2.83%
Total			−566,487		

Across this group, the mean unreported net reduction in incomplete pathways was approximately 2.83% per month. This figure is consistent with the approximately 3% unreported removal rate previously reported across all provider types for the period April 2023 to March 2025 (Watson and Fisher 2025).

In absolute terms, these 22 Large Acute Trusts collectively recorded approximately 566,000 unreported pathway removals during 2024.

Mid and South Essex NHS Foundation Trust

Applying the same methodology to Mid and South Essex NHS Foundation Trust over the same reporting period produces materially different results, as shown in Table 3.

Table 3

	Observed Incomplete Pathways	Expected Incomplete Pathways	Unreported Change	Treatable Pathways	Unreported %
2024-01	160,406	179,690	-19,284	204,932	-9.41%
2024-02	163,701	179,792	-16,091	203,871	-7.89%
2024-03	164,111	185,356	-21,245	209,022	-10.16%
2024-04	164,343	179,804	-15,461	204,266	-7.57%
2024-05	165,697	183,501	-17,804	209,360	-8.50%
2024-06	168,084	184,244	-16,160	208,737	-7.74%
2024-07	163,119	185,758	-22,639	212,808	-10.64%
2024-08	169,019	181,456	-12,437	204,352	-6.09%
2024-09	166,363	184,667	-18,304	209,632	-8.73%
2024-10	164,086	183,887	-19,801	209,184	-9.47%
2024-11	163,517	182,637	-19,120	207,378	-9.22%
2024-12	163,596	180,658	-17,062	200,748	-8.50%

	Observed Incomplete Pathways	Expected Incomplete Pathways	Unreported Change	Treatable Pathways	Unreported %
Mean	164,670	182,621	−17,951	207,024	−8.67%
Total			−215,408		

Across 2024, MSEFT exhibits a mean unreported removal rate of 8.67% of incomplete pathways per month—approximately three times the Large Acute Trust baseline (8.67% vs 2.83%). In absolute terms, this corresponds to 215,408 unreported pathway removals during the year.

Although MSEFT represents only one of the 23 Large Acute Trusts included in this analysis, its unreported removals account for approximately 27.5% of the total unreported removals observed across all Large Acute Trusts in 2024.

Excess unreported removals

To control for national effects such as validation exercises, seasonal pressures, or system-wide disruptions (e.g. winter capacity constraints), a period-by-period comparison was conducted between MSEFT and the Large Acute Trust baseline excluding MSEFT.

For each month, the excess unreported removal rate was calculated as the difference between MSEFT's unreported percentage loss and the corresponding baseline percentage loss among other Large Acute Trusts. Applying this differential to MSEFT's total pathways yields an estimate of excess unreported removals attributable to deviation from the baseline.

Table 5

Period	MSEFT Unreported Change	MSEFT Unreported %	Other Unreported %	Excess Unreported %	Excess Unreported Loss
2024-01	-19,284	-9.41%	-3.05%	-6.36%	-11,434.17
2024-02	-16,091	-7.89%	-2.99%	-4.90%	-8,812.38
2024-03	-21,245	-10.16%	-2.81%	-7.35%	-13,626.98
2024-04	-15,461	-7.57%	-2.68%	-4.89%	-8,790.94
2024-05	-17,804	-8.50%	-2.77%	-5.74%	-10,527.01
2024-06	-16,160	-7.74%	-2.54%	-5.20%	-9,584.01
2024-07	-22,639	-10.64%	-3.18%	-7.45%	-13,847.34
2024-08	-12,437	-6.09%	-2.81%	-3.28%	-5,948.65
2024-09	-18,304	-8.73%	-2.52%	-6.21%	-11,463.41
2024-10	-19,801	-9.47%	-3.05%	-6.41%	-11,793.33
2024-11	-19,120	-9.22%	-2.88%	-6.34%	-11,580.66
2024-12	-17,062	-8.50%	-2.65%	-5.85%	-10,565.05

	MSEFT Unreported Change	MSEFT Unreported %	Other Unreported %	Excess Unreported %	Excess Unreported Loss
Mean	−17,951	−8.66%	−2.83%	−5.83%	−10,664
Total	−215,408				−127,974

Under this comparison, MSEFT recorded 127,974 unreported pathway removals in excess of what would be expected given the observed Large Acute Trust baseline in 2024.

Interpretation and implications

Individual RTT pathway removals may be clinically, administratively, or operationally justified, and this analysis does not assess the validity of individual clock-stop decisions. However, the scale, persistence, and concentration of unreported removals observed at MSEFT—particularly when compared with otherwise similar Large Acute Trusts over the same period—raise questions about the consistency and transparency of RTT waiting list management practices.

Further analysis is planned to examine pathway-level mechanisms and alternative explanatory hypotheses.

At this scale, even a small proportion of affected pathways corresponding to unresolved or abandoned care would translate into significant clinical risk for patients who were referred precisely because investigation or treatment was deemed necessary. Nonetheless, the magnitude of the discrepancies identified here is sufficient, even at this preliminary stage, to warrant closer scrutiny.

References

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